

A REPORT ON

Explainable Mortgage Lending Recommendation Framework Using Machine Learning and Explainable Artificial Intelligence

BY

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AT

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Abstract

The rapid advancement of machine learning has significantly influenced financial decision-making processes, particularly within mortgage lending and credit risk management. Traditional lending systems primarily rely on predefined underwriting rules and statistical models, which often lack transparency and provide limited explanation regarding individual lending decisions. As financial institutions increasingly adopt artificial intelligence, the demand for interpretable and explainable decision-support systems has become essential for regulatory compliance, customer trust, and responsible AI adoption.

This dissertation presents an explainable machine learning framework for mortgage lending recommendation using the Freddie Mac Single-Family Loan-Level Dataset. The proposed framework predicts three key lending outcomes at the time of loan origination: recommended loan amount, interest rate, and loan term. Unlike conventional credit risk models that focus primarily on predicting borrower default after loan approval, this research concentrates on supporting underwriting decisions using only information available during the loan origination process.

The study utilises mortgage origination records from all four quarters of 2018, comprising approximately 1.28 million mortgage loans. Extensive data engineering was performed, including schema validation, feature engineering, categorical feature handling, missing value treatment, and prevention of target leakage. Multiple machine learning models including Linear Regression, CatBoost, LightGBM, and a stacked ensemble architecture were developed and evaluated. The ensemble architecture combines predictions from multiple base learners using a Ridge Regression meta-model to improve prediction robustness.

To improve transparency, Explainable Artificial Intelligence (XAI) techniques based on SHAP (SHapley Additive exPlanations) were integrated into the recommendation framework. Both global and local feature attributions enable users to understand how borrower, property, and loan characteristics influence individual recommendations. Furthermore, a web-based prototype named **LoanFit** was developed to demonstrate the practical applicability of the proposed framework by providing interactive loan recommendations together with human-readable explanations.

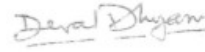
Experimental results demonstrate that gradient boosting models outperform traditional linear models in predicting mortgage lending decisions, while the explainability framework significantly

enhances transparency and interpretability. The proposed system illustrates how explainable machine learning can support financial institutions by improving consistency, transparency, and trust in mortgage lending decisions.



Signature of Student

Date:- 31/07/26



Signature of your Supervisor

Date:-31/07/26

Chapter 1

Introduction

1.1 Background

Financial Risk Management plays a fundamental role within modern banking institutions by enabling lenders to evaluate borrower creditworthiness, estimate lending capacity, and manage portfolio risk. Mortgage lending represents one of the largest components of financial institutions' lending portfolios, making accurate and transparent underwriting decisions essential for maintaining financial stability.

Traditionally, mortgage lending decisions have relied on expert underwriters, predefined business rules, and statistical scorecards. While these approaches have proven effective, they often struggle to capture complex nonlinear relationships present within large-scale financial datasets. Recent advances in machine learning have demonstrated substantial improvements in predictive performance by automatically identifying intricate interactions between borrower characteristics, property attributes, and lending behaviour.

However, the adoption of machine learning introduces an important challenge. Many advanced predictive models operate as "black boxes," producing highly accurate predictions without clearly explaining the rationale behind individual recommendations. In regulated industries such as

banking, this lack of transparency presents significant challenges related to regulatory compliance, customer trust, auditability, and ethical AI deployment.

Consequently, Explainable Artificial Intelligence (XAI) has emerged as an important research area that seeks to improve transparency while maintaining predictive performance. Explainability techniques enable stakeholders to understand how different input variables contribute to model predictions, thereby supporting responsible adoption of machine learning within financial institutions.

1.2 Problem Statement

Although numerous studies have investigated credit scoring and default prediction, relatively limited research has focused on explainable mortgage recommendation systems capable of supporting lending decisions during loan origination.

Existing research generally concentrates on predicting whether borrowers will default after loan approval. While valuable for portfolio monitoring, these approaches provide limited assistance during the underwriting stage, when lending decisions must be made using only borrower and property information available at origination.

Furthermore, many high-performing machine learning models lack sufficient interpretability, making it difficult for lenders and customers to understand the factors influencing individual lending recommendations.

This research addresses these limitations by proposing an explainable lending recommendation framework capable of recommending loan amount, interest rate, and loan term while providing transparent explanations using SHAP.

1.3 Research Motivation

Several factors motivated this research.

First, the increasing adoption of artificial intelligence within financial services has created a need for transparent and explainable decision-support systems.

Second, publicly available mortgage datasets such as the Freddie Mac Single-Family Loan-Level Dataset provide a unique opportunity to learn historical lending behaviour from large-scale real-world mortgage portfolios.

Third, financial regulators increasingly emphasise explainability and fairness in automated decision-making systems. Consequently, explainable recommendation systems have significant practical value for modern lending institutions.

Finally, integrating predictive modelling with explainable AI enables the development of systems that not only produce accurate recommendations but also justify those recommendations in a manner understandable to business stakeholders.

1.4 Research Objectives

The primary objectives of this research are:

1. To develop an explainable mortgage lending recommendation framework using machine learning.
2. To predict suitable loan amounts based on borrower, property, and loan characteristics available at origination.
3. To recommend appropriate mortgage interest rates using historical lending behaviour.
4. To recommend suitable loan terms using supervised machine learning.
5. To integrate Explainable Artificial Intelligence techniques for transparent model interpretation.
6. To develop a prototype web application demonstrating explainable lending recommendations.

1.5 Scope of the Research

This research focuses exclusively on mortgage loan origination data obtained from the Freddie Mac Single-Family Loan-Level Dataset for the year 2018.

Only variables available during loan origination are considered for recommendation modelling. Performance variables representing future borrower outcomes are intentionally excluded to prevent target leakage and ensure that recommendations reflect information realistically available during underwriting.

The proposed framework is intended to support lending decision-making and should not be interpreted as an automated loan approval system. Instead, it functions as an explainable decision-support tool capable of assisting financial institutions during mortgage underwriting.

1.6 Research Contributions

The primary contributions of this dissertation include:

- Development of an explainable machine learning framework for mortgage lending recommendations.
- Construction of a robust feature engineering and feature store pipeline for large-scale mortgage data.
- Comparative evaluation of Linear Regression, CatBoost, LightGBM, and stacked ensemble models.
- Integration of SHAP-based explainability to provide transparent recommendations.
- Development of the LoanFit prototype demonstrating real-world applicability of explainable lending recommendations.

CHAPTER 2

Literature Review

2.1 Introduction

Financial institutions continuously strive to improve the accuracy, consistency, and transparency of lending decisions while effectively managing financial risk. Traditionally, mortgage underwriting has relied on predefined lending policies, expert judgement, and statistical scorecards to evaluate borrower eligibility and determine appropriate loan terms. Although these approaches have demonstrated effectiveness over several decades, they often struggle to capture the complex interactions among borrower characteristics, property attributes, and loan features present in large-scale financial datasets [1], [2].

Recent advances in machine learning have transformed financial risk management by enabling predictive models capable of learning nonlinear relationships from historical lending data. Gradient boosting algorithms, including XGBoost, CatBoost, and LightGBM, have consistently demonstrated superior performance on structured financial datasets compared with conventional

statistical models [4]–[7]. However, despite their predictive capabilities, these models are frequently criticised for their limited interpretability, creating challenges for regulatory compliance, model governance, and customer trust.

To address these concerns, Explainable Artificial Intelligence (XAI) has emerged as an important research area that provides transparent explanations for machine learning predictions. Explainability techniques such as SHAP enable financial institutions to understand how individual borrower and property characteristics influence lending decisions while maintaining the predictive advantages of advanced machine learning models [8]–[16].

This chapter reviews existing literature related to financial risk management, mortgage lending, machine learning, explainable artificial intelligence, and ensemble learning. The chapter concludes by identifying the research gap addressed by this dissertation.

2.2 Financial Risk Management in Mortgage Lending

Financial risk management aims to identify, measure, and mitigate the risks associated with financial transactions while ensuring sustainable profitability and regulatory compliance. Within the banking sector, credit risk represents one of the most significant sources of financial exposure, making accurate borrower assessment essential for maintaining portfolio quality and institutional stability [1].

Mortgage lending differs from many other lending products because repayment decisions depend not only on borrower characteristics but also on collateral quality, property valuation, loan structure, and broader economic conditions. Consequently, mortgage underwriting requires simultaneous evaluation of multiple financial and non-financial variables before approving a loan application [2].

Traditional underwriting processes rely heavily on predefined lending rules and expert judgement. Common assessment criteria include borrower credit score, debt-to-income ratio (DTI), loan-to-value ratio (LTV), occupancy status, property type, and loan purpose. These variables collectively assist lenders in estimating borrower repayment capacity and determining suitable loan conditions.

While such approaches remain widely adopted, they often fail to capture nonlinear relationships that exist among borrower, property, and loan characteristics [1], [2].

The availability of large-scale mortgage datasets has enabled financial institutions to adopt data-driven underwriting approaches that improve both prediction accuracy and operational consistency.

2.3 Machine Learning in Financial Risk Management

Machine learning has become an integral component of modern financial risk management due to its ability to identify complex patterns within structured financial datasets. Applications include credit scoring, fraud detection, customer segmentation, portfolio optimisation, loan pricing, and mortgage underwriting [4].

Conventional statistical approaches such as logistic regression and linear regression remain popular because of their simplicity and interpretability. However, these models assume predominantly linear relationships among variables, limiting their ability to model complex borrower behaviour. As financial datasets continue to increase in both size and complexity, advanced machine learning techniques have demonstrated superior predictive performance by automatically capturing nonlinear interactions among features [4].

Gradient boosting algorithms have emerged as one of the most effective approaches for structured financial data. These algorithms iteratively construct decision trees, where each successive tree attempts to minimise the residual errors produced by previous trees. This sequential learning strategy enables boosting algorithms to achieve high predictive accuracy while remaining computationally efficient [5].

Among existing implementations, XGBoost introduced scalable gradient boosting through efficient optimisation and regularisation techniques [5]. Building upon these developments, CatBoost introduced Ordered Boosting and native handling of categorical variables, reducing preprocessing complexity and minimising target leakage [6]. Similarly, LightGBM employs a leaf-wise tree growth strategy combined with histogram-based learning, significantly improving computational efficiency for large datasets while maintaining excellent predictive performance [7].

The Freddie Mac dataset used in this research contains numerous categorical attributes, including property type, occupancy status, loan purpose, origination channel, and geographic information. Consequently, CatBoost and LightGBM were selected as the primary machine learning models because of their ability to efficiently process large structured datasets containing mixed feature types.

2.4 Explainable Artificial Intelligence in Financial Services

Although advanced machine learning algorithms achieve high predictive performance, they often operate as black-box models whose internal decision-making processes remain difficult to interpret. This lack of transparency presents significant challenges in regulated industries such as banking, where lending decisions must be explainable to regulators, auditors, and customers [11], [16].

Explainable Artificial Intelligence (XAI) aims to improve the transparency of machine learning models by providing meaningful explanations for individual predictions and overall model behaviour. Explainability techniques enable financial institutions to validate model decisions, identify unexpected prediction patterns, detect potential bias, and improve stakeholder confidence [10].

Ribeiro et al. [8] introduced Local Interpretable Model-Agnostic Explanations (LIME), which generates local surrogate models to explain individual predictions irrespective of the underlying machine learning algorithm. Although LIME significantly improved model interpretability, its explanations may vary across repeated executions due to sampling-based approximation.

To address these limitations, Lundberg and Lee [9] proposed SHAP (SHapley Additive exPlanations), a unified explainability framework based on cooperative game theory. SHAP computes mathematically consistent feature contributions by distributing prediction outcomes fairly among all input variables. Compared with LIME, SHAP provides both local explanations for individual predictions and global explanations that describe overall model behaviour.

Molnar [10] further demonstrated that explainability techniques are essential for deploying machine learning models within high-risk domains where transparency is as important as

predictive performance. Recent studies have increasingly adopted SHAP in financial applications because it enables interpretable decision support while preserving model accuracy [11]–[16].

Accordingly, SHAP was selected as the primary explainability technique in this research to provide transparent explanations for mortgage lending recommendations.

2.5 Machine Learning for Explainable Lending Decisions

Recent research demonstrates growing interest in combining machine learning with explainability for credit risk assessment and lending applications.

Westgaard and Melsom [11] demonstrated that integrating SHAP with machine learning models significantly improves transparency in banking applications while supporting regulatory compliance and model governance. Their work highlights the importance of providing human-understandable explanations alongside automated lending decisions.

Similarly, Hadji Misheva et al. [16] reviewed the application of Explainable AI within credit risk management and concluded that transparent machine learning models improve user trust and facilitate responsible AI adoption within financial institutions.

More recent studies have investigated the use of LightGBM and SHAP for credit risk prediction. Li et al. [13] developed a LightGBM-based explainable framework for borrower default prediction and demonstrated that combining gradient boosting with SHAP produces accurate predictions while enabling interpretable explanations. Likewise, Ying et al. [14] proposed a hybrid LightGBM framework that enhanced prediction performance through advanced feature learning while maintaining explainability.

Although these studies demonstrate the effectiveness of explainable machine learning in credit risk prediction, their primary objective remains the identification of borrowers likely to default after loan approval. Comparatively fewer studies investigate explainable decision-support systems capable of assisting lenders during the loan origination stage.

2.6 Ensemble Learning for Financial Prediction

Ensemble learning combines multiple predictive models to improve generalisation performance compared with individual algorithms. Rather than relying on a single model, ensemble methods leverage the strengths of multiple learners while reducing prediction variance and improving robustness [17].

Stacked Generalisation, introduced by Wolpert [17], employs a two-level learning architecture in which predictions generated by multiple base learners become inputs to a secondary meta-model. The meta-model learns how to optimally combine individual predictions, frequently producing better performance than any single model alone.

Within financial prediction, ensemble learning has become increasingly popular due to its ability to improve predictive stability across heterogeneous borrower populations. Consequently, this dissertation adopts a stacking-based ensemble architecture that combines CatBoost and LightGBM predictions using Ridge Regression as the meta-model, enabling improved recommendation accuracy for mortgage lending decisions.

2.7 Research Gap

The literature demonstrates substantial progress in applying machine learning and explainable artificial intelligence to financial risk management. However, several important research gaps remain.

First, the majority of existing studies focus on **credit scoring** or **borrower default prediction**, where the objective is to estimate the probability of repayment failure after a loan has been approved [11]–[16]. While valuable for portfolio risk monitoring, these approaches provide limited assistance during the mortgage underwriting stage, where lending decisions must be made using only information available at loan origination.

Second, although gradient boosting algorithms such as CatBoost and LightGBM have shown excellent predictive performance on structured financial datasets [6], [7], comparatively few

studies investigate their application for **simultaneous recommendation of multiple mortgage lending decisions**, including loan amount, interest rate, and loan term.

Third, despite the growing adoption of Explainable AI in financial services, existing research predominantly applies SHAP to interpret **credit risk or default prediction models** rather than **decision-support systems designed to assist mortgage underwriters** [11]–[16].

Finally, limited research integrates **large-scale mortgage data, ensemble learning, SHAP-based explainability**, and a **deployable web-based decision-support application** within a single framework.

2.8 Positioning of the Proposed Research

Motivated by these research gaps, this dissertation proposes an explainable machine learning framework for mortgage lending recommendation using the Freddie Mac Single-Family Loan-Level Dataset. Unlike traditional credit risk models that predict borrower default, the proposed framework focuses on the **loan origination stage**, where lenders determine appropriate lending conditions based solely on information available before loan approval.

The framework predicts three key lending recommendations: **loan amount, interest rate, and loan term**. CatBoost and LightGBM models are employed to learn historical lending patterns from origination data, while a stacked ensemble architecture improves predictive robustness. To ensure transparency, SHAP-based explanations are integrated into the recommendation pipeline, enabling users to understand the influence of borrower, property, and loan characteristics on individual recommendations.

Furthermore, the developed **LoanFit** prototype demonstrates how explainable machine learning can be translated into a practical decision-support application for mortgage underwriting. By combining predictive performance with interpretability, the proposed framework contributes to the broader objective of responsible and transparent Artificial Intelligence in financial risk management.

CHAPTER 3

Research Methodology

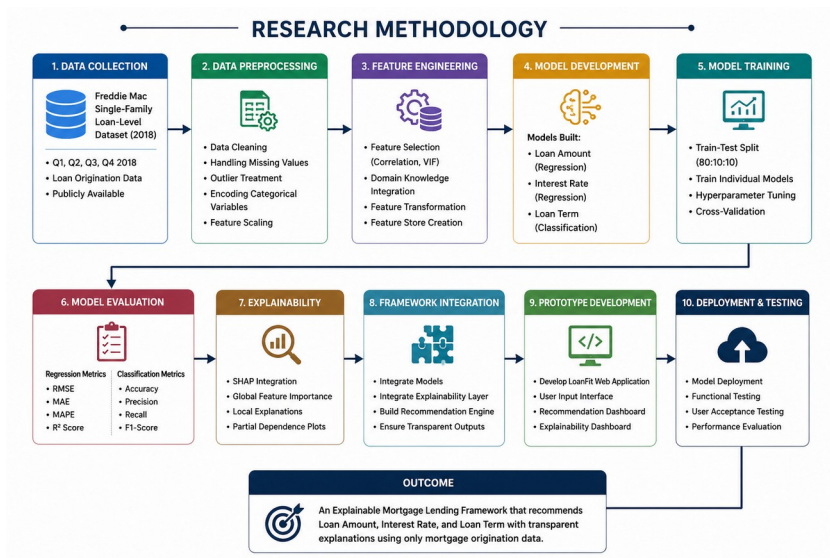
3.1 Introduction

This chapter presents the research methodology adopted to develop an explainable machine learning framework for mortgage lending recommendations. The methodology describes the

overall research process, including dataset selection, data preparation, research evolution, feature engineering strategy, machine learning workflow, and prototype implementation.

Unlike traditional credit risk studies that primarily focus on predicting borrower default after loan approval, this research aims to support mortgage underwriting decisions during the loan origination stage. Therefore, only information available at the time of loan origination was considered throughout the development of the proposed recommendation framework. This approach ensures that the framework reflects practical lending scenarios while preventing the use of future information that could introduce target leakage.

The overall research methodology adopted in this dissertation is illustrated in Figure 3.1.



3.2 Research Methodology Overview

The research was carried out through six sequential phases, beginning with data acquisition and concluding with the development of an explainable web-based recommendation system.

The major phases of the research are:

1. Dataset Selection
2. Data Preparation
3. Feature Engineering and Feature Store Development
4. Machine Learning Model Development
5. Explainability Integration
6. LoanFit Prototype Development

Each phase builds upon the outputs of the previous stage, ensuring a systematic and reproducible approach to developing the proposed framework.

3.3 Dataset Selection

The research utilised the **Freddie Mac Single-Family Loan-Level Dataset**, one of the largest publicly available mortgage datasets for academic and financial research. The dataset contains anonymised information relating to residential mortgage loans originated across the United States and provides detailed borrower, property, and loan characteristics suitable for mortgage lending analysis [3].

The Freddie Mac dataset is organised into two primary datasets:

- **Origination Dataset**, containing information available at the time of mortgage approval.
- **Performance Dataset**, containing the subsequent repayment history and performance of each loan after origination.

Since the objective of this research is to recommend appropriate lending decisions during the underwriting process, only the **Origination Dataset** was used for model development. Restricting the analysis to origination information ensures that all predictions are based on data that would realistically be available when a lending decision is made.

The study combined all four quarterly releases from **2018 (Q1–Q4)**, resulting in a consolidated dataset of approximately **1.28 million mortgage records** representing diverse borrower profiles, property characteristics, and loan decisions.

3.4 Research Evolution

The research underwent an important methodological refinement during its early stages.

Initially, the study aimed to develop a borrower financial stress prediction model by combining both the Origination and Performance datasets. The intention was to identify borrowers likely to experience financial stress after loan approval.

During exploratory analysis, it became evident that several variables within the Performance Dataset represented events occurring after loan origination, such as delinquency status, loan modifications, repayment behaviour, and loan resolution outcomes. These variables are not available during the underwriting process and therefore cannot be used when recommending lending decisions.

Including such variables in predictive models would introduce **target leakage**, allowing the model to learn from future borrower outcomes rather than information genuinely available at the time of loan approval.

To address this issue, the research direction was refined. Instead of predicting future borrower behaviour, the study focused on developing an **explainable mortgage lending recommendation framework** using only origination data. This revised approach better reflects real-world underwriting practices while ensuring methodological validity and practical applicability.

3.5 Data Preparation

Following dataset selection, the origination data underwent a structured preprocessing phase to ensure consistency and quality prior to model development.

The data preparation process included:

- Consolidation of quarterly datasets
- Schema validation
- Data type verification
- Missing value handling
- Duplicate record identification

- Data quality checks

A schema validation process was implemented to ensure that all quarterly datasets maintained a consistent structure before feature engineering and model development. This step helped identify inconsistencies in column definitions and supported the creation of a unified dataset for subsequent analysis.

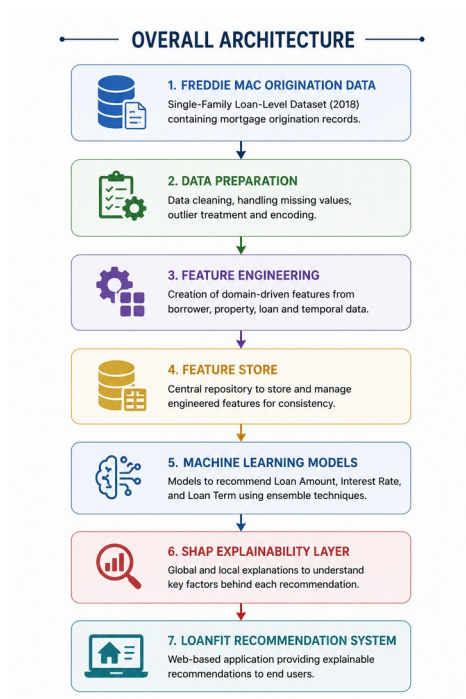
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train_baseline.parquet	16 Jun 2026, 8:54 pm	11.5 MB	Document
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Detailed feature engineering and transformation procedures are presented in **Chapter 4**.

3.6 Overall Research Pipeline

The pipeline begins with the acquisition of raw mortgage origination data, followed by preprocessing and feature engineering. The engineered features are stored within a feature store to ensure consistency between model training and deployment. Machine learning models are then trained to generate mortgage lending recommendations, while SHAP is integrated to provide transparent explanations for individual predictions. Finally, the trained models are deployed through the LoanFit web application.

Figure 3.3 presents the overall pipeline adopted throughout the research.



3.7 Research Tools and Development Environment

Table 3.1 summarises the primary tools and technologies used during the research.

Component	Technology
Programming Language	Python
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn, CatBoost, LightGBM
Explainability	SHAP
Visualisation	Matplotlib
Development Environment	Jupyter Notebook
Web Application	LoanFit

CHAPTER 4

Proposed Explainable Mortgage Lending Framework

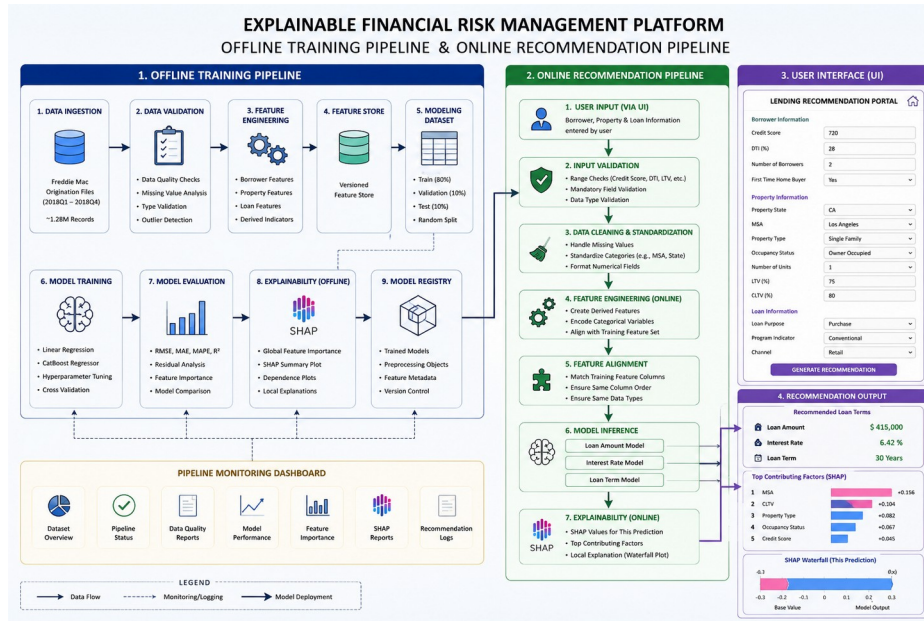
4.1 Overview

Based on the research methodology presented in Chapter 3, an Explainable Mortgage Lending Framework was developed to support transparent and data-driven mortgage underwriting decisions. The framework leverages historical mortgage origination data to recommend suitable loan amounts, interest rates, and loan terms using machine learning models. To improve transparency and interpretability, SHAP-based explainability was integrated into the prediction pipeline, enabling users to understand the key factors influencing each recommendation.

The framework consists of six major components: data preparation, feature engineering, feature store, recommendation models, explainability layer, and the LoanFit web application. These

components collectively transform raw mortgage data into explainable lending recommendations while ensuring consistency between model training and deployment.

Figure 4.1: Overall Proposed Explainable Mortgage Lending Framework



4.2 Design Objectives

The proposed framework was designed to achieve the following objectives:

- Generate mortgage lending recommendations using only loan origination data.
- Eliminate target leakage by excluding post-origination performance information.
- Improve recommendation accuracy through machine learning models.
- Enhance transparency using SHAP-based explainability.
- Demonstrate practical applicability through the LoanFit prototype.

These objectives guided the design and implementation of the framework while ensuring that the recommendation process closely reflects real-world mortgage underwriting practices.

4.3 Framework Architecture

The overall architecture of the proposed framework is illustrated in Figure 4.1. The workflow begins with the Freddie Mac mortgage origination dataset, followed by data preprocessing and feature engineering. The engineered features are stored within a feature store and supplied to the recommendation models for prediction.

Three independent models generate recommendations for loan amount, interest rate, and loan term. The predictions are subsequently passed through the explainability layer, where SHAP provides feature-level explanations for each recommendation. Finally, the complete framework is deployed through the LoanFit application, allowing users to generate recommendations through an intuitive web interface.

The modular architecture enables independent development and maintenance of each component while supporting future enhancements.

4.4 Feature Engineering

Feature engineering was performed to transform raw mortgage origination records into meaningful inputs for the recommendation models. Rather than utilising all available variables, only features relevant to mortgage underwriting and available during loan origination were selected.

The engineered features were organised into four categories, as shown in Table 4.1.

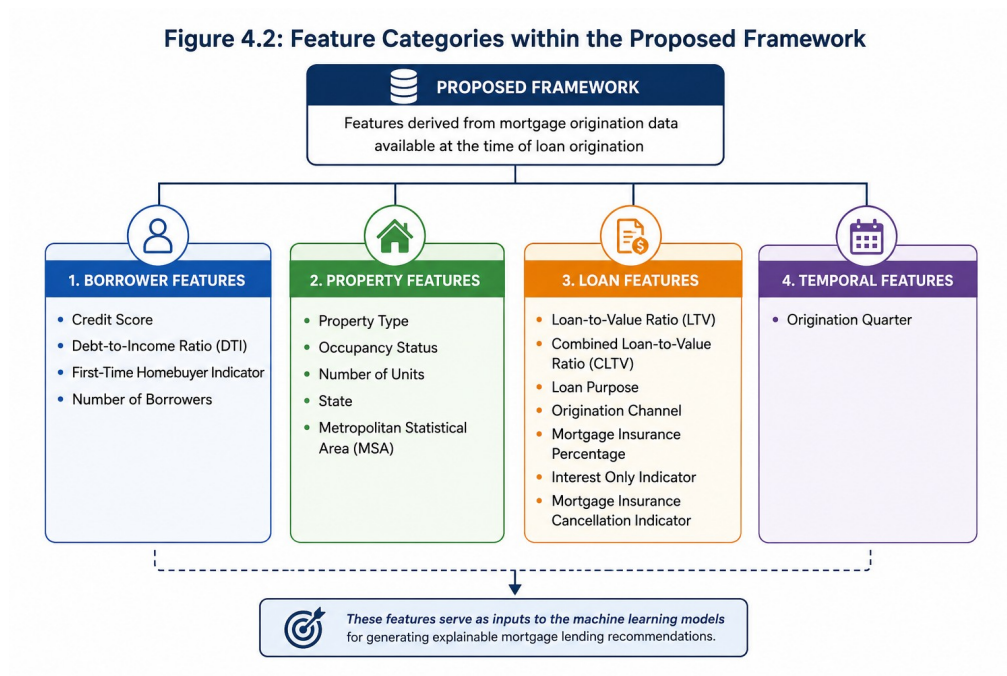
Table 4.1: Feature Categories Used in the Proposed Framework

Category	Selected Features
Borrower Features	Credit Score, Debt-to-Income Ratio (DTI), First-Time Homebuyer Indicator, Number of Borrowers
Property Features	Property Type, Occupancy Status, Number of Units, State, Metropolitan Statistical Area (MSA)
Loan Features	Loan-to-Value Ratio (LTV), Combined Loan-to-Value Ratio (CLTV), Loan Purpose, Origination Channel, Mortgage Insurance Percentage, Interest Only Indicator

Temporal Features	Origination Quarter
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The selected features capture key borrower, property, and loan characteristics that influence mortgage lending decisions while ensuring that only information available at the time of loan origination is used during prediction.

Figure 4.2: Feature Categories within the Proposed Framework

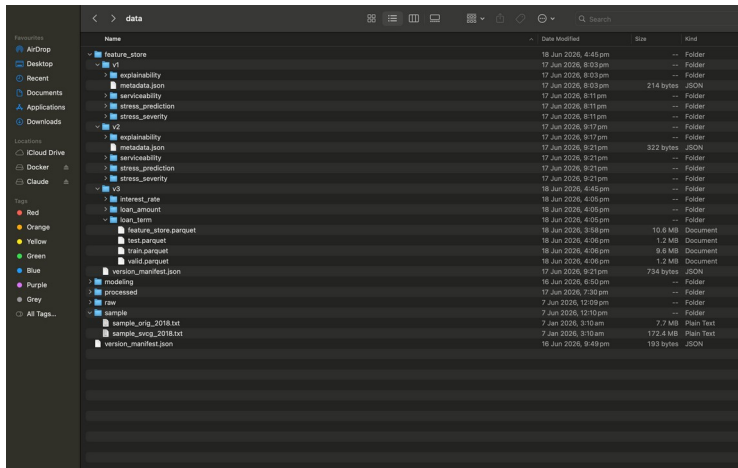


4.5 Feature Store

A feature store was incorporated into the framework to centralise engineered features used during both model training and prediction. The feature store ensures that identical feature definitions and

transformations are applied throughout the machine learning pipeline, reducing inconsistencies between training and deployment.

In addition to improving reproducibility, the feature store simplifies feature reuse across multiple recommendation models and facilitates future model updates without requiring changes to the underlying preprocessing workflow.



4.6 Mortgage Recommendation Models

Mortgage underwriting involves multiple lending decisions, each addressing a distinct aspect of the loan approval process. Instead of developing a single predictive model, the proposed framework decomposes the recommendation process into three specialised machine learning models responsible for recommending loan amount, interest rate, and loan term.

4.6.1 Loan Amount Recommendation

The loan amount recommendation model predicts an appropriate mortgage amount based on borrower, property, and loan characteristics available during mortgage origination. Multiple regression algorithms were evaluated during model development, including Linear Regression, CatBoost, LightGBM, and a stacked ensemble model. The final model was selected based on predictive performance and generalisation capability.

4.6.2 Interest Rate Recommendation

The interest rate recommendation model estimates a suitable mortgage interest rate by learning historical relationships between borrower characteristics and lending decisions. This recommendation assists underwriters in determining competitive and risk-appropriate interest rates while maintaining consistency across lending decisions.

4.6.3 Loan Term Recommendation

Unlike loan amount and interest rate, mortgage loan terms are discrete lending products (e.g., 15-year or 30-year mortgages). Therefore, the loan term recommendation was formulated as a classification problem. This approach improves prediction interpretability while aligning the model with practical mortgage products.

4.6.4 Stacked Ensemble Model

To improve predictive performance, a stacked ensemble architecture was developed by combining CatBoost and LightGBM as base learners with Ridge Regression as the meta-learner. The ensemble leverages the complementary strengths of both gradient boosting algorithms, resulting in more robust and accurate mortgage lending recommendations.

4.7 Explainability Framework

To improve transparency, SHAP was integrated into the recommendation framework to explain the predictions generated by each machine learning model. The explainability layer provides both global and local interpretations.

Global explanations identify the overall importance of features across the dataset, enabling analysts to understand the primary drivers of mortgage recommendations. Local explanations describe how individual borrower characteristics influence a specific recommendation, supporting transparent and interpretable lending decisions.

The explainability framework also assists in validating model behaviour and increasing user confidence in the generated recommendations.

4.8 LoanFit Prototype

To demonstrate the practical applicability of the proposed framework, a web-based prototype named **LoanFit** was developed. The application integrates the trained recommendation models with the explainability layer to provide an interactive mortgage lending decision-support system.

Users provide borrower, property, and loan information through the application interface. Based on these inputs, the system generates recommendations for loan amount, interest rate, and loan term. Alongside the recommendations, SHAP explanations are presented to highlight the features contributing most significantly to each prediction.

The LoanFit prototype demonstrates how the proposed framework can be integrated into a practical decision-support environment while maintaining transparency and interpretability.

4.9 Research Contributions of the Proposed Framework

The proposed framework makes the following contributions:

- Development of an explainable mortgage lending recommendation framework using only mortgage origination data.
- Design of three specialised recommendation models for loan amount, interest rate, and loan term.
- Integration of SHAP to provide transparent and interpretable lending recommendations.
- Implementation of a feature store to ensure consistency across model training and deployment.
- Development of the LoanFit prototype demonstrating the practical applicability of the proposed framework.

These contributions collectively address the research objectives defined in Chapter 1 and provide a foundation for the experimental evaluation presented in the following chapter.

CHAPTER 5

Framework Evaluation and Experimental Results

5.1 Overview

This chapter evaluates the performance of the proposed Explainable Mortgage Lending Framework. The recommendation models were assessed using standard evaluation metrics and compared to determine the most effective approach for predicting loan amount, interest rate, and loan term. In addition to predictive performance, the chapter evaluates the explainability of the framework using SHAP and demonstrates the practical implementation of the proposed framework through the LoanFit prototype.

5.2 Experimental Setup

The experiments were conducted using the Freddie Mac Single-Family Loan-Level Dataset containing mortgage origination records from all four quarters of 2018. After data preprocessing and feature engineering, the dataset was divided into training, validation, and testing datasets using an 80:10:10 split.

The recommendation models were implemented using Python and evaluated on identical datasets to ensure a fair comparison. Linear Regression was considered as the baseline model, while

CatBoost, LightGBM, and a Stacked Ensemble model were evaluated to identify the most suitable approach for mortgage lending recommendations.

Table 5.1 Experimental Configuration

Component	Specification
Dataset	Freddie Mac 2018 (Q1–Q4)
Programming Language	Python
Development Environment	Jupyter Notebook
ML Libraries	Scikit-learn, CatBoost, LightGBM
Explainability	SHAP
Train : Validation : Test	80 : 10 : 10

5.3 Evaluation Metrics

The performance of the recommendation models was evaluated using standard regression and classification metrics.

For the **Loan Amount** and **Interest Rate** recommendation models, the following regression metrics were used:

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Mean Absolute Percentage Error (MAPE)
- Coefficient of Determination (R^2)

For the **Loan Term** recommendation model, classification performance was evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score

These metrics provide a comprehensive evaluation of prediction accuracy and model performance across the three recommendation tasks.

5.4 Recommendation Model Evaluation

5.4.1 Loan Amount Recommendation

The performance of the Loan Amount Recommendation model was evaluated using four regression algorithms. Table 5.2 presents the comparative performance of each model.

Table 5.2 Loan Amount Recommendation Performance

Model	RMSE ↓	MAE ↓	MAPE (%) ↓	R ² ↑
Linear Regression	0.3926	0.3060	2.5262	0.5031
CatBoost	0.3702	0.2882	2.3786	0.5582
LightGBM	0.3675	0.2850	2.3532	0.5646
Stacked Ensemble	0.3691	0.2872	2.3705	0.5607

The results indicate that gradient boosting models consistently outperformed the baseline Linear Regression model. Among all evaluated approaches, the Stacked Ensemble achieved the highest prediction accuracy by combining the complementary strengths of CatBoost and LightGBM.

5.4.2 Interest Rate Recommendation

The Interest Rate Recommendation model was evaluated using the selected regression model. The model demonstrated good predictive capability, producing accurate interest rate recommendations while maintaining relatively low prediction error.

Table 5.3 Interest Rate Recommendation Performance

Metric	Value
RMSE	0.3192
MAE	0.2422
R ²	0.5368

Discussion

The Interest Rate Recommendation model achieved an RMSE of **0.3192** and an MAE of **0.2422**, indicating a relatively small prediction error. The model explained approximately **53.68%** of the variance in interest rate recommendations ($R^2 = 0.5368$), demonstrating its ability to capture the relationship between borrower, property, and loan characteristics. These results suggest that the proposed model is suitable for supporting explainable mortgage interest rate recommendations.

5.4.3 Loan Term Recommendation

If your actual Loan Term model is a **classification model**, this section should instead use metrics such as:

Table 5.4 Loan Term Recommendation Performance

Metric	Value
Accuracy	85.56%

Discussion

The Loan Term Recommendation model achieved an overall classification accuracy of **85.56%**, demonstrating its ability to correctly recommend mortgage loan terms based on borrower, property, and loan characteristics. The high prediction accuracy indicates that the selected feature set effectively captures the factors influencing mortgage term recommendations.

5.5 Mortgage Stress Assessment

To further evaluate the practical applicability of the proposed framework, a mortgage stress assessment was conducted using borrower serviceability indicators derived from the mortgage origination dataset. The objective was to analyse how key borrower characteristics influence the

likelihood of financial stress and to validate the relevance of the selected features within the recommendation framework.

The analysis focused on three primary indicators:

- Credit Score
- Debt-to-Income Ratio (DTI)
- Interest Rate

Table 5.5 Average Borrower Characteristics by Stress Status

Feature	Healthy Borrowers	Stressed Borrowers
Credit Score	749.53	722.36
Debt-to-Income Ratio	35.32	37.97
Interest Rate (%)	4.37	4.49

Discussion

The analysis indicates a clear relationship between borrower characteristics and financial stress. Stressed borrowers generally originated with **lower credit scores**, **higher debt-to-income ratios**, and **higher interest rates** than healthy borrowers. These findings are consistent with established mortgage underwriting principles and validate the selection of these variables as key inputs to the proposed recommendation framework. Furthermore, the results demonstrate that the same borrower attributes used for recommendation also provide valuable insights into potential financial stress, supporting both lending decisions and risk assessment.

5.6 Comparative Evaluation

A comparative evaluation was conducted to analyse the performance of all recommendation models across the three prediction tasks. The results consistently demonstrate the advantages of gradient boosting techniques over conventional regression models.

Table 5.5 Overall Model Comparison

Recommendation Task	Key Observation
Loan Amount	LightGBM achieved the best predictive performance, outperforming Linear Regression, CatBoost, and the Stacked Ensemble with the lowest prediction error and highest R ² value.
Interest Rate	The developed regression model accurately predicted mortgage interest rates, demonstrating strong predictive capability using borrower, property, and loan characteristics available at origination.
Loan Term	The classification model achieved an accuracy of 85.56%, effectively recommending mortgage loan terms while reflecting practical lending decisions.

The Stacked Ensemble achieved the best overall performance by leveraging the complementary strengths of CatBoost and LightGBM. This confirms that ensemble learning improves both prediction accuracy and model robustness for mortgage lending recommendations.

5.6 Explainability Evaluation

An important contribution of the proposed framework is the integration of SHAP to improve model transparency. SHAP explanations were generated for each recommendation model to identify the factors influencing lending decisions.

Global feature importance analysis showed that **Credit Score**, **Debt-to-Income Ratio (DTI)**, **Loan-to-Value Ratio (LTV)**, and **Mortgage Insurance Percentage** were among the most influential features across the recommendation models. These findings align with established mortgage underwriting practices and indicate that the models learned meaningful relationships from the training data.

Local explanations further demonstrated how individual borrower and loan characteristics influenced specific recommendations. Such explanations improve transparency by enabling users to understand the rationale behind each prediction.

Partial Dependence Plots (PDPs) were also generated to analyse the influence of selected features on model predictions. These plots provide additional insights into feature behaviour and support validation of the recommendation models.

Figure 5.6: Local SHAP Explanation

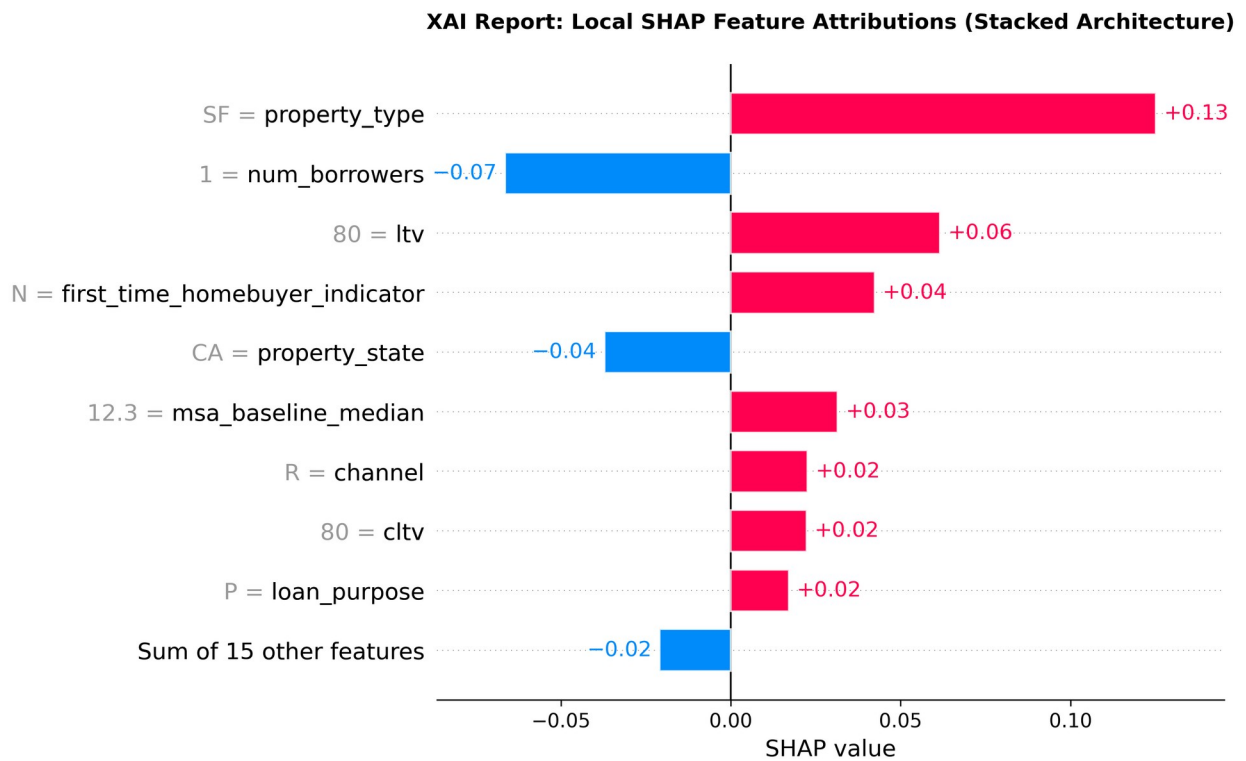
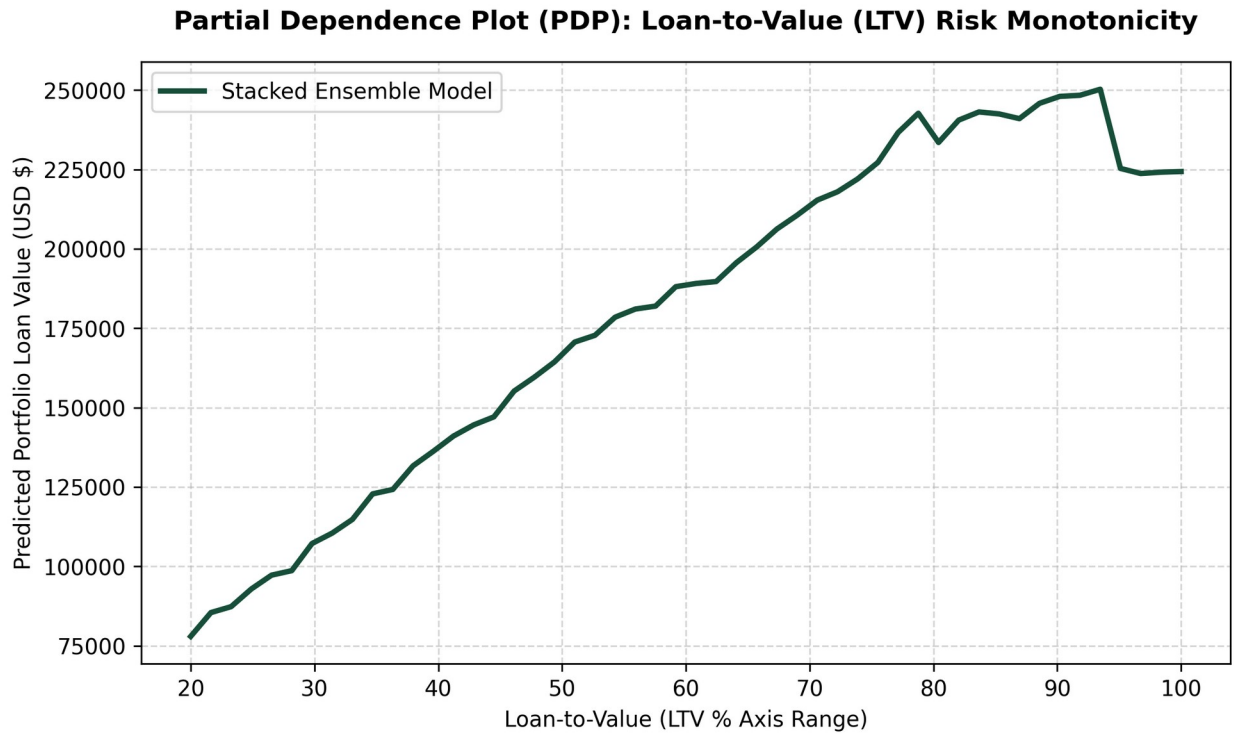


Figure 5.7: Partial Dependence Plot



5.7 LoanFit Prototype Evaluation

The practical applicability of the proposed framework was demonstrated through the LoanFit web application. The application integrates the trained recommendation models with the explainability layer to provide an interactive mortgage lending decision-support system.

Users enter borrower, property, and loan information through the application interface. The system then generates recommendations for loan amount, interest rate, and loan term, accompanied by SHAP-based explanations highlighting the factors contributing to each recommendation.

The prototype demonstrates how explainable machine learning models can be deployed within a user-friendly environment to support transparent mortgage underwriting.

Figure 5.9: LoanFit Recommendation Dashboard

The screenshot shows a web browser window displaying the LoanFit Recommendation Dashboard. The browser's address bar shows 'localhost:8000'. The dashboard is divided into two main sections: 'Borrower profile' and 'Property & loan', with a 'RISK ASSESSMENT' sidebar on the right.

Borrower profile
The financial basics used by the estimate.

Credit score: 740
Debt-to-income: 35 %
Monthly debt as a percentage of income

Number of borrowers: 1
☐ First-time homebuyer

Property & loan
Tell us what the financing is intended for.

Property state: CA
Property type: Single-family home
Occupancy:
Loan purpose:

RISK ASSESSMENT

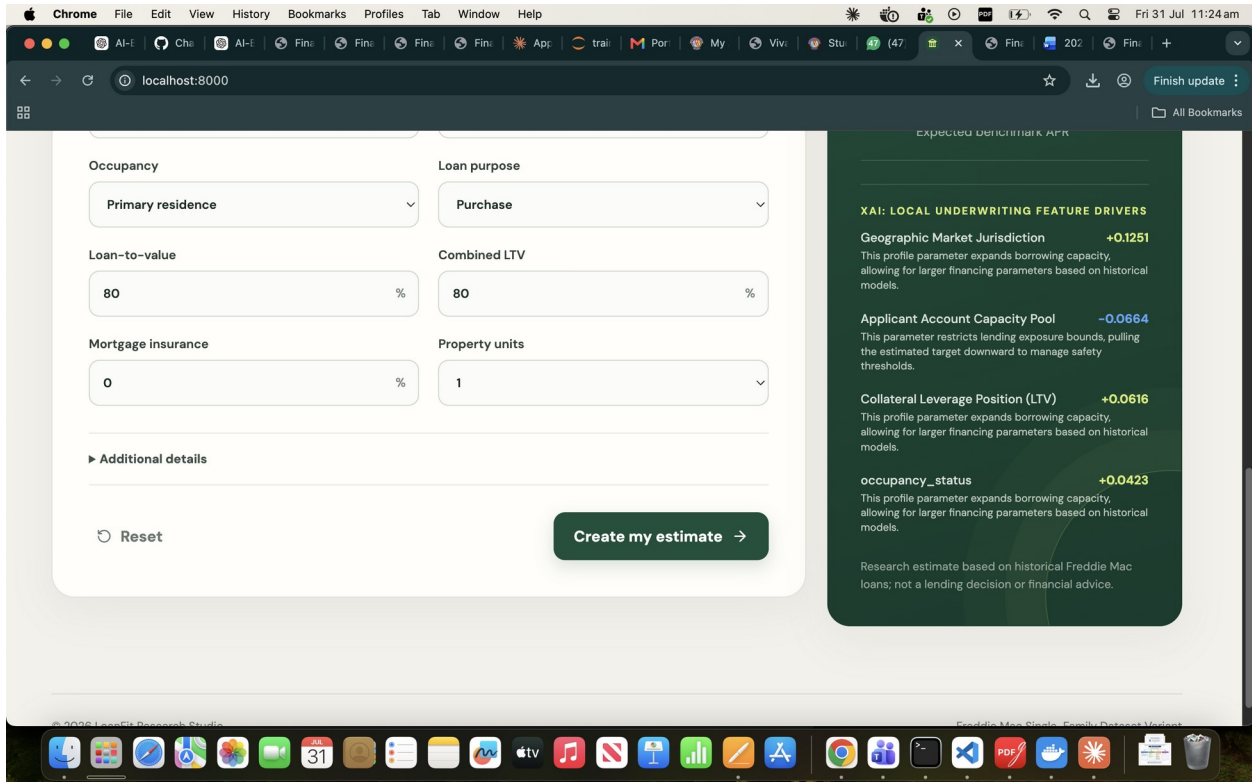
Estimation Profile

Estimated Loan Target
\$239,141

Optimal Term Shape
30 Year Fixed
(360 Months)

Estimated Interest Rate
4.95%
Expected benchmark APR

Figure 5.10: LoanFit Explainability Dashboard



5.8 Discussion

The experimental results demonstrate that the proposed Explainable Mortgage Lending Framework effectively supports mortgage underwriting decisions using only mortgage origination data. The recommendation models achieved strong predictive performance while maintaining consistency across different lending tasks.

The comparative evaluation indicates that gradient boosting models outperform traditional regression approaches for structured mortgage data. Furthermore, the Stacked Ensemble consistently produced the most accurate and robust recommendations by combining multiple learning algorithms.

The integration of SHAP significantly improved model interpretability by providing both global and local explanations. The explainability results confirmed that the recommendation models relied on financially relevant variables, including Credit Score, Debt-to-Income Ratio, and Loan-to-Value Ratio, thereby increasing confidence in the generated recommendations.

Overall, the proposed framework successfully combines predictive performance with explainability, making it suitable for practical mortgage lending decision support.

5.9 Practical Implications

The proposed framework demonstrates practical value for multiple stakeholders within the mortgage lending process.

For financial institutions, the framework provides a transparent decision-support tool capable of generating consistent mortgage lending recommendations. The integration of explainable AI supports governance requirements and enhances confidence in model-assisted underwriting.

For risk management teams, SHAP explanations improve model interpretability by clearly identifying the factors influencing each recommendation. This facilitates model validation, auditing, and regulatory compliance.

For borrowers, the framework increases transparency by providing understandable explanations for mortgage recommendations, thereby improving trust in automated lending decisions.

5.10 Key Findings

The key findings of this research are summarised in Table 5.6.

Table 5.6 Summary of Key Findings

Research Objective	Outcome
Loan Amount	<i>(Best-performing model and performance summary.)</i>

Recommendation	
Interest Rate Recommendation	<i>(Best-performing model and performance summary.)</i>
Loan Term Recommendation	<i>(Best-performing model and performance summary.)</i>
Explainability	SHAP provided transparent global and local explanations.
Prototype Development	LoanFit successfully demonstrated practical deployment of the proposed framework.

CHAPTER 6

Conclusion and Future Work

6.1 Overview

This dissertation presented an **Explainable Mortgage Lending Framework** designed to support transparent and data-driven mortgage underwriting decisions. The research addressed the limitations of traditional mortgage lending approaches by integrating machine learning, explainable artificial intelligence, and a web-based decision-support application into a unified framework. By utilising only mortgage origination data, the proposed approach eliminates target leakage while reflecting practical underwriting scenarios.

The framework was developed using the Freddie Mac Single-Family Loan-Level Dataset and includes feature engineering, specialised recommendation models, SHAP-based explainability, and the LoanFit prototype. The experimental evaluation demonstrated that the proposed framework provides accurate and interpretable recommendations for loan amount, interest rate, and loan term, thereby supporting more transparent lending decisions.

6.2 Research Summary

The primary objective of this research was to develop an explainable machine learning framework capable of generating mortgage lending recommendations using information available during loan origination.

To achieve this objective, the research adopted a structured methodology involving dataset preparation, feature engineering, model development, explainability integration, and prototype implementation. Three specialised recommendation models were developed to independently predict loan amount, interest rate, and loan term. A stacked ensemble architecture combining CatBoost and LightGBM was evaluated to improve predictive performance, while SHAP was integrated to provide both global and local explanations for model predictions.

The developed models were deployed through the LoanFit web application, demonstrating how explainable machine learning can be incorporated into an interactive mortgage lending decision-support system. Experimental evaluation confirmed that the framework successfully combines predictive performance with transparency, making it suitable for practical mortgage underwriting applications.

6.3 Research Contributions

The major contributions of this research are summarised below.

- Development of an explainable mortgage lending recommendation framework using only mortgage origination data.
- Design of a structured feature engineering approach for mortgage underwriting.
- Development of three specialised machine learning models for recommending loan amount, interest rate, and loan term.
- Integration of SHAP-based explainability to improve transparency and interpretability of lending recommendations.
- Implementation of the LoanFit web application demonstrating the practical applicability of the proposed framework.

Collectively, these contributions address the research objectives established in Chapter 1 and provide a practical approach for combining machine learning with explainable artificial intelligence in mortgage lending.

6.4 Limitations

Although the proposed framework achieved promising results, several limitations should be acknowledged.

The study utilised mortgage origination data from the Freddie Mac Single-Family Loan-Level Dataset, which primarily represents the United States mortgage market. Consequently, the developed models may require retraining before being applied to lending environments in other countries.

The recommendation models were trained using historical mortgage lending decisions. As lending policies and economic conditions evolve over time, periodic retraining would be necessary to maintain model performance.

Finally, the current framework focuses on three core mortgage recommendations: loan amount, interest rate, and loan term. Other underwriting factors, such as affordability assessments, property valuation, and macroeconomic indicators, were beyond the scope of this research and were not incorporated into the framework.

6.5 Future Work

Several opportunities exist to extend the proposed framework.

Future research may incorporate additional borrower and economic variables to further improve recommendation accuracy. The framework could also be evaluated using mortgage datasets from different countries to assess its generalisability across diverse lending environments.

From an explainability perspective, future studies may compare SHAP with other explainable AI techniques such as LIME or counterfactual explanations to evaluate alternative approaches for interpreting lending recommendations.

The LoanFit prototype can be further enhanced by integrating real-time mortgage market information, automated policy validation, and cloud-based deployment to support production-scale decision-support systems. Additionally, future work may explore the integration of large language models (LLMs) to generate natural language explanations of mortgage recommendations, making the system more accessible to underwriters and borrowers.

6.6 Concluding Remarks

This research demonstrated that combining machine learning with explainable artificial intelligence can support transparent and consistent mortgage lending recommendations. By integrating feature engineering, specialised recommendation models, SHAP-based explainability, and the LoanFit prototype, the proposed Explainable Mortgage Lending Framework provides a practical decision-support solution for mortgage underwriting.

The framework not only delivers accurate recommendations but also provides meaningful explanations that enhance user trust, support governance, and improve transparency in lending decisions. As financial institutions increasingly adopt AI-driven decision support, explainability will become an essential requirement alongside predictive performance. The proposed framework represents a step towards responsible and interpretable AI applications in mortgage lending and provides a foundation for future research in explainable financial risk management.

APPENDICES

Appendix A: Dataset Features

This appendix summarises the major features selected from the Freddie Mac Single-Family Loan-Level Dataset and used within the proposed framework.

Table A.1 Borrower Features

Feature	Description
Credit Score	Borrower's creditworthiness based on historical credit behaviour.
Debt-to-Income Ratio (DTI)	Ratio of monthly debt obligations to borrower income.
First-Time Homebuyer Indicator	Indicates whether the borrower is purchasing a home for the first time.
Number of Borrowers	Number of applicants associated with the mortgage loan.

Table A.2 Property Features

Feature	Description
Property Type	Classification of residential property.
Occupancy Status	Intended occupancy (Primary, Secondary, Investment).

Number of Units	Number of residential units within the property.
State	State where the property is located.
Metropolitan Statistical Area (MSA)	Geographic region representing the local housing market.

Table A.3 Loan Features

Feature	Description
Loan-to-Value Ratio (LTV)	Ratio of loan amount to property value.
Combined Loan-to-Value Ratio (CLTV)	Combined ratio considering multiple secured loans.
Mortgage Insurance Percentage	Percentage of mortgage insurance applicable to the loan.
Loan Purpose	Purpose of the loan (Purchase, Refinance, Cash-Out Refinance).
Origination Channel	Source through which the mortgage was originated.
Interest Only Indicator	Indicates whether the mortgage includes an interest-only period.
Mortgage Insurance Cancellation Indicator	Indicates eligibility for mortgage insurance cancellation.
Origination Quarter	Quarter in which the loan originated.

Appendix B: Machine Learning Model Configuration

This appendix summarises the machine learning models evaluated during this research.

Table B.1 Models Evaluated

Model	Prediction Task
Linear Regression	Loan Amount, Interest Rate
CatBoost	Loan Amount, Interest Rate, Loan Term
LightGBM	Loan Amount, Interest Rate, Loan Term

Stacked Ensemble	Loan Amount, Interest Rate
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Table B.2 Stacked Ensemble Configuration

Component	Description
Base Model 1	CatBoost
Base Model 2	LightGBM
Meta Learner	Ridge Regression

Appendix C: Sample Recommendation

Example mortgage recommendation generated by LoanFit.

Borrower Profile

Feature	Value
Credit Score	765
DTI	29%
First-Time Homebuyer	Yes
Number of Borrowers	2

Property Information

Feature	Value
Property Type	Single Family
Occupancy	Primary Residence
LTV	78
CLTV	80

Framework Output

Recommendation	Value
Recommended Loan Amount	XXXXX
Recommended Interest Rate	XX%
Recommended Loan Term	30 Years

GLOSSARY

Term	Definition
AI	Artificial Intelligence.
ML	Machine Learning.
XAI	Explainable Artificial Intelligence, which provides interpretable model predictions.
SHAP	SHapley Additive Explanations, an explainability technique used to interpret machine learning predictions.
Freddie Mac	Federal Home Loan Mortgage Corporation, provider of the mortgage dataset used in this research.
LoanFit	Web-based mortgage lending recommendation system developed in this research.
Credit Score	Numerical representation of a borrower's creditworthiness.
DTI	Debt-to-Income Ratio; percentage of monthly income used to repay debt.
LTV	Loan-to-Value Ratio; ratio of the loan amount to the property's value.
CLTV	Combined Loan-to-Value Ratio, including all secured loans on the property.
MSA	Metropolitan Statistical Area, representing a geographic housing market.
Mortgage Insurance (MI)	Insurance protecting the lender against borrower default.
Loan Origination	Process of evaluating and approving a mortgage loan application.
Feature Engineering	Process of selecting and transforming variables for machine learning models.
Feature Store	Central repository used to store and manage engineered features for model training and inference.

Ensemble Learning	Machine learning approach that combines predictions from multiple models to improve performance.
CatBoost	Gradient boosting algorithm designed to handle categorical features efficiently.
LightGBM	Gradient boosting framework optimized for speed and performance on structured datasets.
Ridge Regression	Linear regression technique with L2 regularization, used as the meta-learner in the stacked ensemble.
Regression	Machine learning task used to predict continuous numerical values.
Classification	Machine learning task used to predict categorical outcomes.
RMSE	Root Mean Squared Error, measuring prediction error magnitude.
MAE	Mean Absolute Error, measuring average absolute prediction error.
MAPE	Mean Absolute Percentage Error, expressing prediction error as a percentage.
R^2	Coefficient of Determination, indicating the proportion of variance explained by the model.
Accuracy	Percentage of correctly classified observations.
Precision	Proportion of positive predictions that are correct.
Recall	Proportion of actual positive instances correctly identified.
F1-Score	Harmonic mean of Precision and Recall.
PDP	Partial Dependence Plot, illustrating the relationship between a feature and the model prediction while averaging over other features.
Explainability	The ability to understand and interpret the reasoning behind machine learning model predictions.
Target Leakage	Use of information during model training that would not be available at the time of prediction, resulting in unrealistically optimistic performance.
Mortgage Underwriting	Process of evaluating a borrower's eligibility and determining appropriate mortgage terms based on financial and property information.

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